# Measurement and Computation Fusion Methods for Electromagnetic Environment Analysis in Urban Areas

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***Abstract*—The rapid development of wireless communication devices has led to an increasingly complex electromagnetic environment in urban areas, making traditional methods insufficient for accurate and rapid predictions of electromagnetic field distributions. This paper proposes a method for urban electromagnetic environment characterization based on measurement and computation fusion. Initially, the static electromagnetic field distribution is obtained using a collaborative solution algorithm that combines the Method of Moments (MoM) and the Uniform Geometrical Theory of Diffraction (UTD). Then, real-time field values are captured using a limited number of electromagnetic signal measurement devices. By fusing these measurements with simulations, accurate real-time electromagnetic field distributions are achieved. Experimental results demonstrate that this method accurately predicts the electromagnetic field distribution in a 4 km² area of Xi'an, with an error of less than 8 dB between predicted and measured values.**

***Index Terms*—Electromagnetic field distribution; Fusion of measurement and computation; Ray tracing method; urban environment**

## I. Introduction

In recent years, various prediction models for electromagnetic wave propagation in urban environments have been developed. Among these, the Ray Tracing Method (RTM) [1] combined with the UTD [2] is widely used for predicting electromagnetic wave distribution in cities. While UTD effectively handles propagation characteristics in urban streets, it is a high-frequency approximation algorithm that cannot accurately simulate complex radiation sources, leading to lower accuracy in large-scale urban electromagnetic environment analysis. To address this issue, scholars have proposed hybrid computational methods based on both high and low frequency strategies, utilizing precise full-wave algorithms such as the Method of Moments (MoM) to handle complex radiation sources. These methods also use high-frequency approximation algorithms, such as Physical Optics (MOM-PO) [3], Shooting and Bouncing Rays (MOM-SBR) [4], and MOM-UTD [5], to handle challenges like urban buildings. The MoM-UTD algorithm improves computational accuracy and saves resources by using MoM to calculate the radiation field of complex sources and using this field as the incident field for UTD rays. However, the complexity of urban environment modeling and the long computational time for ray tracing still limit real-time performance.

To achieve rapid and accurate analysis of electromagnetic field distributions in urban environments, this paper proposes a measurement and computation fusion method for electromagnetic environment characterization. This approach combines UTD for ray tracing and obtaining information on reflection and diffraction points, employs a parallel computing strategy based on MPI, and applies the MoM[6] for precise calculation of complex antenna radiation characteristics. It then uses UTD to calculate the static electromagnetic field distribution. By deploying electromagnetic signal measurement devices to obtain real-time field strength values and applying the measurement and computation fusion method to correct and update the static field online, this approach enables real-time and accurate characterization of the electromagnetic environment in large-scale urban areas.

## II. Algorithm Principle

### A. *MPI-Based MoM-UTD Hybrid Algorithm*

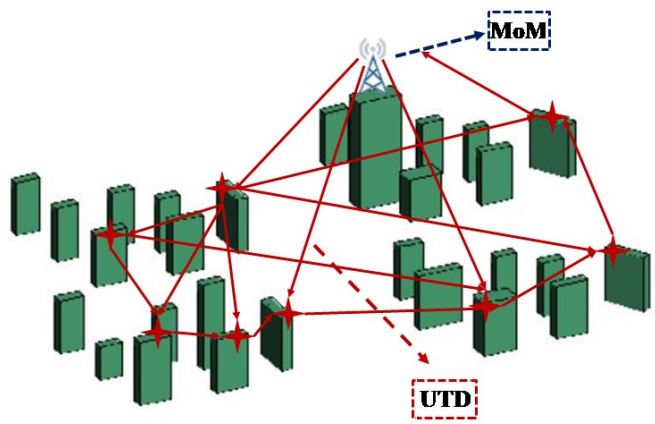


Fig. 1. Schematic Diagram of the MoM-UTD Algorithm

In order to describe the principle of the MPI-based MoM-UTD hybrid algorithm, an antenna and building, as shown in Figure 1, are selected. First, UTD is used for ray tracing to obtain the coordinates of the reflection and diffraction points. Since the contribution of higher-order rays to the field strength at the observation points is very small, and considering the large computational cost of ray tracing in urban environments, only first-order and second-order rays are considered.

MoM utilizes bilinear patches to subdivide complex antennas, allowing for a more accurate approximation of the surface

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current distribution of the antenna. The current on the bilinear patches can be represented as:

$$\boldsymbol{J}_s(p,s)=\sum_{i=0}^{N_p}\sum_{j=0}^{N_s}a_{ij}\boldsymbol{F}_{ij}(p,s)\,. \tag{1}$$

Next, the radiation field calculated by MoM is used as the initial incident field at the reflection and diffraction points. Using the UTD method, the reflection and diffraction fields at these points can be calculated as:

$$\begin{cases}\left[\boldsymbol{E}^r\right]=[R]\left[\boldsymbol{E}^i\right]f\left(s^r\right)\\ \left[\boldsymbol{E}^d\right]=[D]\left[\boldsymbol{E}^i\right]f\left(s^d\right)\end{cases}. \tag{2}$$

Where the matrices for reflection and diffraction coefficients are represented by $[R]$ and $[D]$ respectively.

Analysis reveals that the computational load for the electromagnetic field in urban environments primarily comes from the ray tracing part, which accounts for approximately 97% of the total computation time.

### *B. Electromagnetic data correction technology based on measurement and computation fusion*

This paper adopts an electromagnetic data correction technique based on measurement and computation fusion. By using a small number of electromagnetic signal measurement devices to obtain measured data, the simulation data for urban environments can be corrected.

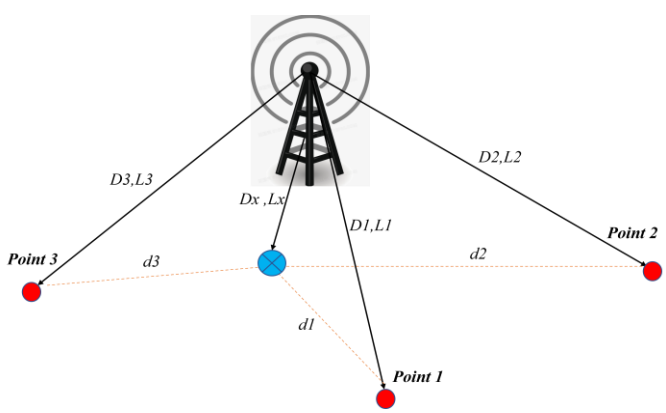


Fig. 2. Electromagnetic Data Correction Principle Based on Measurement and computation Fusion

As shown in Figure 2, let the number of electromagnetic signal measurement devices in the environment be n, with the measured values given by P={p1, p2, ⋯, pn}, and the corresponding simulation values at the measurement points given by S={s1, s2,⋯, sn}. For the next predicted point n+1 as shown in Figure 2, its simulation value is sn+1. Based on the simulation values and measured values of nearby points, its predicted value $p_{n+1}$ can be calculated as:

$$p_{n+1}=s_{n+1}+\Delta p_n\,. \tag{3}$$

Where $\Delta p_n$ represents the correction amount for the simulation values based on the first n measured values:

$$\Delta p_n=\frac{\sum_{i=1}^{m}\omega_i\bullet(p_i-s_i)}{\sum_{i=1}^{m}\omega_i}\,. \tag{4}$$

In this equation, $\omega_i$ is a weighting factor influenced by both power and distance.

Fig. 3. Geographic Model of the 4 Square Kilometer Urban Environment of Xi'an

For the next predicted point $n$+2, its predicted value $p_{n+2}$ can be calculated as:

$$p_{n+2}=s_{n+2}+\Delta p_{n+1}\,. \tag{5}$$

This process is repeated to traverse all observation points in the area to be tested, calculating the field strength values for each observation point.

## III. Results and Discussion

To validate the effectiveness of the proposed measurement and computation fusion method for urban electromagnetic environment analysis, a simulation and experimental analysis of the 4 square kilometers urban electromagnetic environment of Xi'an is conducted.

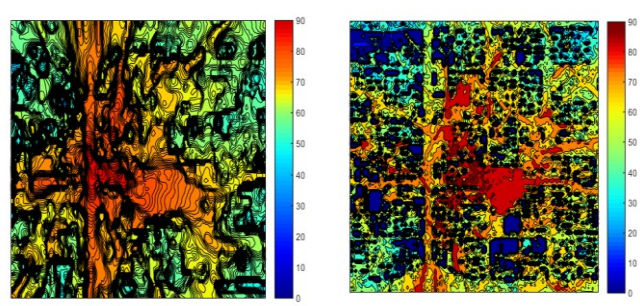

Fig. 4. Simulated and measured values of electromagnetic fields in urban environments

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For the 4 square kilometer urban environment of Xi'an, a precise geographic model was created, as shown in Figure 3. The radiation source has an emission frequency of 409.5 MHz. A supercomputing cluster was used to perform simulations with the MPI-based MoM-UTD hybrid algorithm, yielding the static distribution of the electromagnetic field in the 4 square kilometer urban environment of Xi'an, as shown in Figure 4.

Then, the proposed measurement and computation fusion technique was used to calibrate the simulation values. Electromagnetic signal measurement devices were deployed in the 4 square kilometer area, obtaining real measured field strengths at the locations of the devices. The proposed method was then applied to make real-time corrections to the static field in the region.

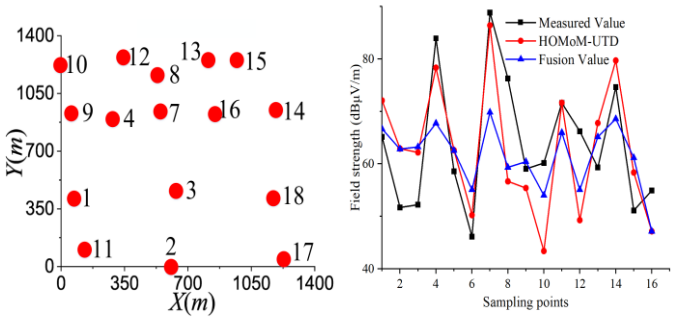


Fig. 5. Test point distribution and comparison of simulation, test and fusion values

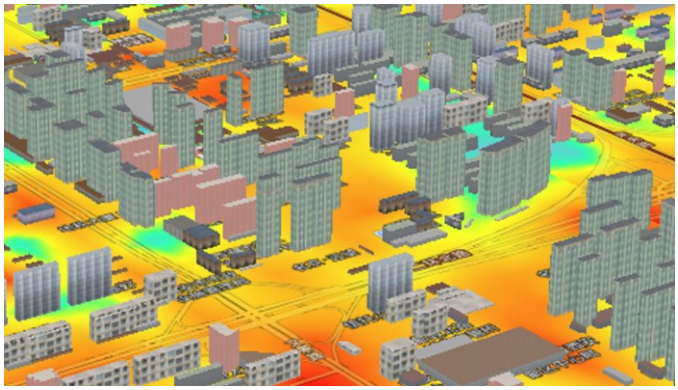

Fig. 6. Distribution of electromagnetic fields in local urban environments

Figure 5 shows the distribution of multiple electromagnetic signal measurement devices in a 4 km² region, and presents a comparison of the simulation, measured, and fused values for this region. From the figure, the root mean square (RMS) error between the simulation and measurement values is 13.99 dB, while the RMS error between the fusion and test values is 7.09 dB.

Figure 6 shows the modified local urban environmental electromagnetic field distribution. Numerical statistics indicate that the method proposed in this paper predicts the electric field values with an RMS error of less than 8 dB compared to the actual measured values.

## IV. Conclusion

This paper presents a simulation of the electromagnetic environment in a 4 square kilometer area of Xi'an, and applies an el This paper presents a simulation of the electromagnetic environment in a 4 square kilometer area of Xi'an and applies an electromagnetic data correction technique based on measurement and computation fusion to adjust the static electromagnetic field obtained from the simulation.

By comparing the simulation, measurement, and fused values of the urban electromagnetic environment, it is evident that the method proposed in this paper offers high accuracy in predicting electromagnetic field distributions in urban environments and demonstrates certain real-time capabilities. This approach not only provides protective monitoring for the urban electromagnetic environment but also serves as a reference for the accuracy verification of electromagnetic wave propagation models.

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